

Supplementary Note on the Dimension Five Lie-Algebra Exclusion

Human-requested explanatory expansion

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Purpose

This note expands the representation-theoretic step in the proposed proof of the five-dimensional case. The point to justify is:

Claim. *There is no compact 5-dimensional Lie subalgebra of $\mathfrak{so}(5)$.*

In the Banach-space application, Auerbach's reduction places $\mathcal{Z}(X)$ inside $\mathfrak{so}(5)$, while Rosenthal's gap theorem leaves $\dim \mathcal{Z}(X) = 5$ as the only possible missing value below the Euclidean case. Thus the above Lie-algebra exclusion is the only local classification step needed for $n = 5$.

Compact Lie Algebras of Dimension Five

Let $\mathfrak{g}$ be a compact real Lie algebra with $\dim \mathfrak{g} = 5$. Compact Lie algebras split as

$$\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus [\mathfrak{g}, \mathfrak{g}],$$

where $[\mathfrak{g}, \mathfrak{g}]$ is compact semisimple. The smallest nonzero compact simple Lie algebra is $\mathfrak{su}(2)$, of real dimension 3. Hence there are only two possibilities:

$$\mathfrak{g} \cong \mathbb{R}^5 \quad \text{or} \quad \mathfrak{g} \cong \mathfrak{su}(2) \oplus \mathbb{R}^2.$$

There is no room in dimension 5 for two nonzero simple factors, since that would already have dimension at least $3 + 3 = 6$.

If $\mathfrak{g}$ is abelian, then an embedding $\mathfrak{g} \hookrightarrow \mathfrak{so}(5)$ is impossible. Indeed, an abelian subalgebra of $\mathfrak{so}(5)$ has dimension at most

$$\text{rank } \mathfrak{so}(5) = 2.$$

Equivalently, a commuting family of real skew-symmetric matrices can be simultaneously put into 2×2 rotation blocks plus a zero line, so at most two independent rotation parameters are possible in dimension 5.

It remains to exclude

$$\mathfrak{g} \cong \mathfrak{su}(2) \oplus \mathbb{R}^2.$$

Small Real $\mathfrak{su}(2)$ -Modules

Suppose that $\mathfrak{su}(2) \oplus \mathbb{R}^2$ embeds into $\mathfrak{so}(5)$. This gives a faithful orthogonal representation on $\mathbb{R}^5$. Restrict it to the $\mathfrak{su}(2)$ -summand.

The irreducible complex representations of $\mathfrak{su}(2)$ are

$$V_m = \text{Sym}^m(\mathbb{C}^2), \quad \dim_{\mathbb{C}} V_m = m + 1.$$

Their real forms are standard:

- if m is even, V_m is of real type and gives a real irreducible module of dimension $m + 1$;
- if m is odd, V_m is of quaternionic type and gives a real irreducible module of dimension $2(m + 1)$.

Therefore the nontrivial irreducible real $\mathfrak{su}(2)$ -modules of dimension at most 5 have dimensions

$$3, \quad 4, \quad 5.$$

More precisely:

real dimension	source	type
3	V_2	real type
4	V_1 as a real module	quaternionic type
5	V_4	real type.

The only other summands of dimension at most 5 are trivial real lines. Since the full representation is faithful, the $\mathfrak{su}(2)$ -summand must act nontrivially. Thus the possible decompositions of $\mathbb{R}^5$, as an $\mathfrak{su}(2)$ -module, are:

$$5, \quad 4 + 1, \quad 3 + 1 + 1.$$

Centralizer Check

The $\mathbb{R}^2$ -summand in

$$\mathfrak{su}(2) \oplus \mathbb{R}^2$$

commutes with $\mathfrak{su}(2)$. Therefore, in any orthogonal representation, its image must lie in the skew centralizer of the $\mathfrak{su}(2)$ -action:

$$\{A \in \mathfrak{so}(5) : [A, \rho(\mathfrak{su}(2))] = 0\}.$$

Since the representation of the whole Lie algebra is faithful, this centralizer must contain a 2-dimensional abelian subalgebra. We now check that this never happens.

Case 5

If $\mathbb{R}^5$ is the irreducible 5-dimensional real-type module, then Schur's lemma gives

$$\text{End}_{\mathfrak{su}(2)}(\mathbb{R}^5) = \mathbb{R}.$$

The only commuting endomorphisms are scalar multiples of the identity. A real scalar multiple of the identity is skew-symmetric only if the scalar is zero. Hence the skew centralizer is

$$0.$$

So the central $\mathbb{R}^2$ cannot act faithfully.

Case 4 + 1

On the 4-dimensional quaternionic irreducible module,

$$\text{End}_{\mathfrak{su}(2)}(\mathbb{R}^4) \cong \mathbb{H}.$$

Inside the orthogonal endomorphisms, the skew centralizer is the imaginary quaternionic part:

$$\mathfrak{sp}(1) \cong \mathfrak{so}(3),$$

of dimension 3. However, $\mathfrak{sp}(1)$ has rank 1: any abelian subalgebra of $\mathfrak{sp}(1)$ has dimension at most 1. Concretely, two imaginary quaternions commute only when they are linearly dependent.

The extra trivial 1-dimensional summand contributes no skew operators, since $\mathfrak{so}(1) = 0$. There are also no off-diagonal commuting blocks between the nontrivial irreducible 4-dimensional module and the trivial line, because the nontrivial module has no invariant vector and no invariant functional.

Thus the whole skew centralizer has abelian rank at most 1, not 2. Again the central $\mathbb{R}^2$ cannot act faithfully.

Case 3 + 1 + 1

On the 3-dimensional irreducible real-type module,

$$\text{End}_{\mathfrak{su}(2)}(\mathbb{R}^3) = \mathbb{R},$$

so the skew centralizer on that block is 0. The remaining 2-dimensional trivial part contributes

$$\mathfrak{so}(2),$$

which is 1-dimensional. As before, there are no off-diagonal commuting blocks from the nontrivial irreducible block to the trivial part, because the 3-dimensional irreducible has no fixed vectors or invariant functionals.

Therefore the skew centralizer is only 1-dimensional, so it cannot contain a faithful image of $\mathbb{R}^2$.

Conclusion

Both possible compact 5-dimensional Lie algebras are excluded:

$$\mathbb{R}^5 \quad \text{cannot embed into } \mathfrak{so}(5) \text{ because } \text{rank } \mathfrak{so}(5) = 2,$$

and

$$\mathfrak{su}(2) \oplus \mathbb{R}^2 \quad \text{cannot embed faithfully into } \mathfrak{so}(5)$$

because the $\mathbb{R}^2$ -summand would need a 2-dimensional abelian action in the skew centralizer of one of the possible $\mathfrak{su}(2)$ -module decompositions, and the centralizer check above shows that no such action exists.

Hence no compact 5-dimensional Lie subalgebra of $\mathfrak{so}(5)$ exists. This supports the dimension-five exclusion used in the proposed solution:

$$\dim \mathcal{Z}(X) \neq 5$$

for a 5-dimensional real Banach space X .